

Backend Engineer (AI Software)

About the Role

We are looking for a backend engineer who can design and build reliable, production-grade services that bring AI to life in consumer-facing products. You will work hands-on with LLMs, RAG pipelines, and agentic workflows to integrate AI capabilities into Masan's Consumer Engagement Platform.

What You Will Do

- Design and build backend services that integrate AI tools and models (LLMs, RAG, AI agents) into consumer-facing products - loyalty, promotion, CRM, and engagement workflows.
- Own end-to-end implementation: API design, data flow, prompt engineering pipelines, AI output validation, and observability.
- Build reliable async processing systems that handle non-deterministic AI outputs without breaking downstream consistency.
- Integrate with AI providers (OpenAI, Gemini, Claude, or similar) and design abstraction layers that make swapping models or providers low-risk.
- Work closely with product and data teams to define how AI features interact with user data, business rules, and downstream systems.
- Write high-quality tests (unit, integration) for AI-integrated code paths, including strategies for testing non-deterministic outputs.
- Set up observability for AI services: latency tracking, token usage, error rates, and output quality monitoring.

Requirements

- 3+ years of backend engineering experience with proficiency in at least one relevant language: Golang, Python, Java, or similar.
- Hands-on experience integrating AI/LLM APIs into production systems - not just prototypes.

- Solid understanding of prompt engineering: context injection, output parsing, structured response handling, and failure modes.
- Experience building async systems: event-driven architecture, worker pools, message queues (Kafka or similar).
- Strong fundamentals in distributed systems: idempotency, retry logic, consistency across service boundaries.
- Comfortable with PostgreSQL, Redis, and vector databases (Pinecone, Weaviate, Qdrant, or similar) for AI use cases.
- Familiar with RAG pipelines: embedding generation, chunking strategies, retrieval and re-ranking.
- Experience with Kubernetes, Docker, and CI/CD for production deployments.
- Observability mindset: structured logging, metrics, distributed tracing.

Nice to Have

- Proficiency in Golang - the team's primary backend language.
- Experience with AI agent frameworks: LangChain, LangGraph, or similar.
- Familiarity with AI workflow orchestration (Temporal or similar) for multi-step AI pipelines.
- Background in consumer-facing products at scale (loyalty, promotion, e-commerce, or fintech).
- Experience evaluating AI model performance and implementing feedback loops for continuous improvement.